Project Navi

Bachelor Designer & Developer

NOAILLES Thomas

PROJECT NAVI

Opportunity Analysis

Context of the Request

The autonomy of people with visual impairments is a major issue in their daily lives. Visually impaired and blind individuals regularly face difficulties when navigating urban environments. Identifying a pedestrian crossing, determining the status of a traffic light, or detecting a temporary obstacle can be significant challenges and may pose risks to their safety.

To compensate for their visual impairment, several solutions already exist, such as the white cane, guide dogs, and various mobile applications designed to improve accessibility. These tools help users gain greater independence while ensuring their safety during travel.

At the same time, recent advances in artificial intelligence, computer vision, and mobile technologies are creating new opportunities for the development of assistive tools capable of analyzing the surrounding environment in real time. Modern smartphones are now equipped with high-performance cameras and sufficient computing power to perform complex processing tasks directly on the device.

In this context, it is relevant to explore the opportunity of developing a mobile application capable of providing additional information about the urban environment in order to assist visually impaired individuals during their daily movements and navigation.

Origin of the Need

Visually impaired and blind individuals must constantly adapt to an environment that is primarily designed for sighted people. Despite the existing assistive solutions, certain situations remain difficult to manage and understand.

For example, a white cane enables users to detect obstacles located in their immediate vicinity but does not provide information about elements that are farther away, such as pedestrian crossings, traffic lights, or obstacles several meters ahead. Guide dogs offer more advanced assistance; however, their acquisition and maintenance involve significant costs and are not accessible to everyone.

In addition, urban environments are constantly changing. Construction work, temporary barriers, detours, parked vehicles, and objects left on sidewalks can all create additional challenges for safe navigation.

The identified need is therefore to provide visually impaired individuals with a complementary assistance tool capable of interpreting key elements of their surroundings and quickly informing them of their presence, thereby improving both their autonomy and safety during travel.

Project Positioning

The project is positioned as a mobility assistance solution designed for visually impaired and blind individuals.

The application is not intended to replace existing assistive devices such as the white cane, guide dogs, or mobility techniques taught to people with visual impairments. Instead, it should be considered a complementary tool that provides additional information about the user's surroundings.

This approach aims to enhance users' autonomy while minimizing the risks associated with excessive reliance on technology.

The application should consistently remind users that it serves as a decision-support tool and not as a system that guarantees absolute safety.

Project Overview

The project consists of developing a mobile application that uses a smartphone's camera to analyze the outdoor environment in real time.

By leveraging computer vision and artificial intelligence algorithms, the application should be capable of identifying various elements that are relevant to mobility, including:

- pedestrian crossings;
- traffic lights;
- obstacles located along the user's path;
- temporary barriers;
- construction zones and detours;
- vehicles that may obstruct movement.

The detected information will be communicated to the user through voice messages and vibration feedback, ensuring accessibility for both visually impaired and blind individuals.

The system should prioritize a simple, fast, and intuitive user interaction in order to minimize the user's cognitive load while navigating their environment.

Expected Benefits

The development of this application could provide several benefits.

For Users

- improved autonomy during daily travel;
- better awareness of the urban environment;
- enhanced sense of safety and confidence;
- access to complementary assistance available at any time.

For Caregivers and Families

- increased confidence in the user's ability to travel independently;
- reduced concerns related to mobility and navigation.

For Specialized Organizations

- availability of an additional support tool;
- complement to the existing solutions offered to visually impaired individuals.

Existing Solutions

White cane

The white cane is the most widely used mobility aid for visually impaired individuals. It allows users to detect obstacles in their immediate environment through physical contact. However, it does not provide information about distant elements such as traffic lights or pedestrian crossings.

Guide dogs
Guide dogs offer advanced assistance by helping users navigate obstacles and move safely in urban environments. However, they require extensive training, regular care, and represent a high financial cost, making them inaccessible to some users.

Mobile accessibility applications
Several mobile applications use computer vision and AI to provide environmental assistance (e.g., object recognition or navigation support). While promising, these solutions often remain limited in real-time performance, accuracy, or contextual understanding.

Summary
Existing solutions each address part of the problem, but none fully provides real-time, comprehensive, and affordable environmental awareness for urban mobility.

Example 1: Seeing AI

- **Strengths:**
    - real-time object recognition
    - text reading (OCR)
    - good accessibility features
- **Weaknesses:**
    - relies heavily on internet/cloud processing
    - limited real-time navigation assistance outdoors

Example 2: Google Lookout

- **Strengths:**
  - object and text detection
  - integrated into Android ecosystem
- **Weaknesses:**
  - limited contextual understanding
  - not optimized for complex urban mobility

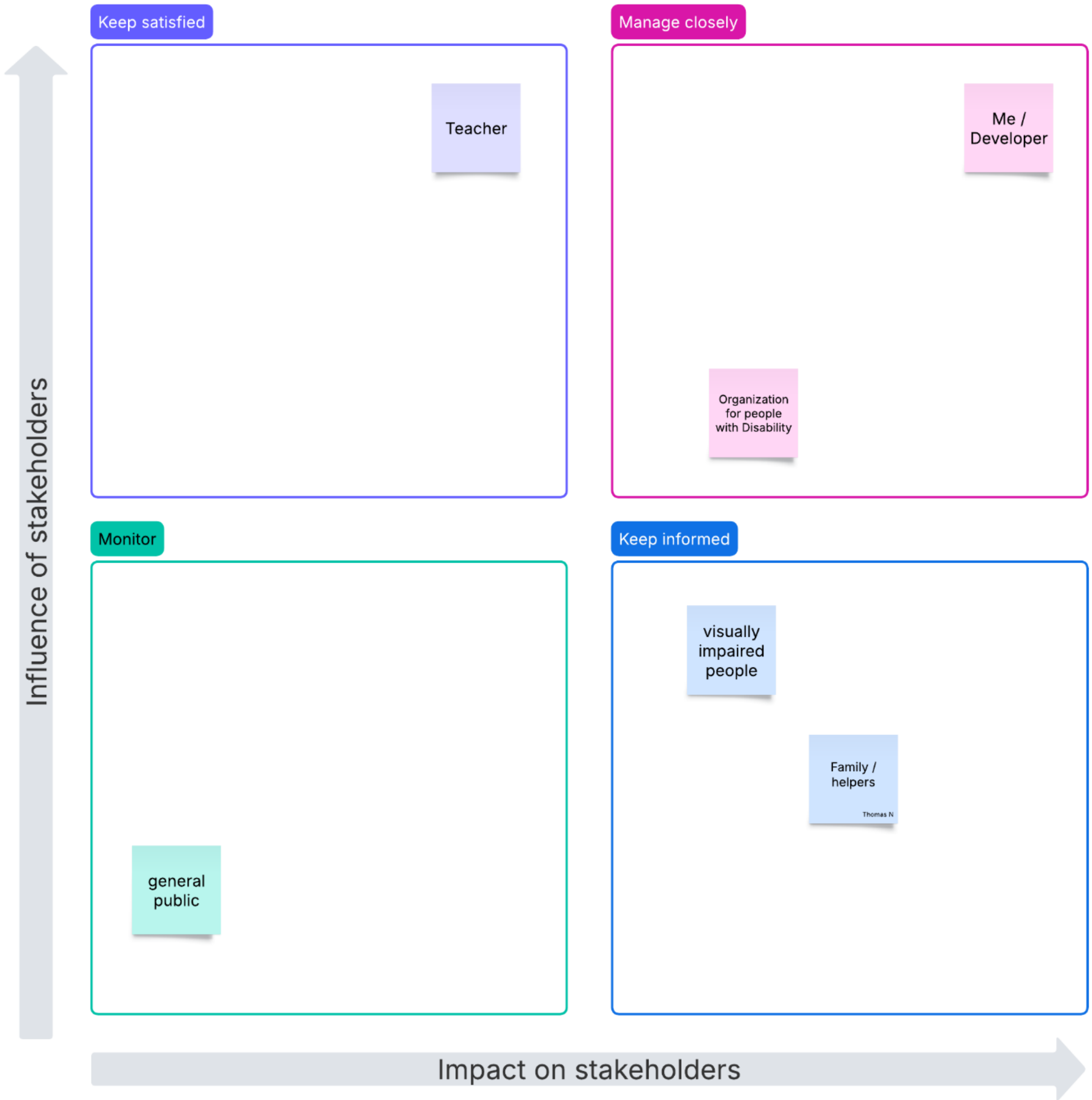
Example 3: Be My Eyes (optional)

- **Strengths:**
  - human assistance available
  - high reliability through volunteers
- **Weaknesses:**
  - not automated (depends on humans)
  - not real-time continuous support

Stakeholders

Stakeholders	Role in the project	Interest
Visually impaired users	Users of the aaplication	Very high
Developer (me)	Design and implementation of the application	Very high
Teacher	Project validation and evaluation	High
Family members / caregivers	Indirect users, support users	High
Organizations for people with disabilities	Advisory role, accessibility guidance	High
General Public	Indirect impact / future potential users	Low

Power Matrix



This classification highlights that the most critical stakeholders are the developer, the academic supervisor, and accessibility organizations, as they have both high influence and high involvement in the project.

End users and caregivers, while highly interested, should be regularly informed to ensure the solution meets their needs.

Information System Context

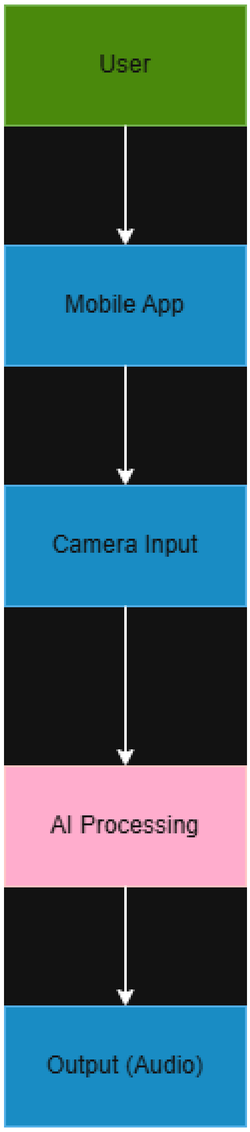
The proposed system is a mobile-based information system designed to assist visually impaired and blind users in their daily mobility. It relies on the interaction between the user, a smartphone device, and embedded software components.

The main actors of the system include:

- the visually impaired user, who interacts with the application and receives feedback;
- the smartphone, which serves as the main hardware platform;
- the mobile application, which processes data and coordinates system functions;
- the smartphone camera, responsible for capturing the surrounding environment;
- the artificial intelligence module, which analyzes images using computer vision techniques;
- the feedback system, which provides information through voice output and vibration signals.

The information flow can be described as follows: the smartphone camera captures real-time images of the environment, which are then processed by the application using AI-based detection algorithms. The system identifies relevant objects such as pedestrian crossings, traffic lights, or obstacles. The interpreted information is then transmitted to the user through audio feedback and haptic signals.

This system aims to transform visual environmental data into accessible information in real time, in order to support safe and autonomous mobility for visually impaired users.



Project Risks and Limitations

The project nevertheless involves several constraints and risks that should be identified during the study phase.

The reliability of object and environment detection represents a major challenge. Incorrect interpretation of the surroundings could lead to inaccurate information, potentially placing the user in a difficult or unsafe situation.

System performance may also be affected by various factors, including:

- low-light conditions;
- adverse weather conditions;
- camera obstruction;
- variations in smartphone hardware quality;
- limited battery life.

Furthermore, the application cannot guarantee perfect detection of all situations encountered in public spaces. For this reason, its use should always be considered complementary to traditional assistive devices and established mobility practices used by visually impaired individuals.

Conclusion

The opportunity analysis highlights a clear need for additional support to improve the autonomy and safety of visually impaired individuals in urban environments. While existing solutions such as white canes and guide dogs provide essential assistance, they remain limited in their ability to deliver real-time environmental awareness.

The proposed mobile application aims to complement these existing tools by using smartphone-based computer vision and artificial intelligence to provide contextual information about the surrounding environment.

The study confirms that the project is technically feasible, although certain limitations such as detection reliability and environmental conditions must be considered.

Overall, this project appears relevant and promising as a complementary solution to enhance accessibility, autonomy, and safety for visually impaired users.

User Analysis

Personas

To better understand the needs, expectations, and challenges of future users, three personas were created. These personas represent the main user groups targeted by the application and guided the design decisions throughout the project.

Main Persona – Lucas

Lucas

age: 21
residence: Creteil
education: Bachelor student
occupation: Administrative Assistant
marital status: University Student

I want to explore new places without having to rely on someone else.

Lucas is a visually impaired university student who is very comfortable with technology. He uses accessibility features on his smartphone daily and relies on public transportation to move around independently. He enjoys discovering new places but often feels stressed when navigating unfamiliar environments.

Comfort With Technology

INTERNET

SOFTWARE

MOBILE APPS

SOCIAL NETWORK

Criteria For Success:

Navigate independently in unfamiliar places.
Avoid unexpected obstacles.
Reach destinations safely and confidently.

Needs

- Real-time obstacle detection.
- Accurate voice guidance.
- Fast and intuitive interactions.

Wants

- Personalized navigation preferences.
- Integration with public transportation information.

Values

- Independence.
- Efficiency.
- Innovation.
- Accessibility.

Fears

- Missing important obstacles.
- Losing orientation in unfamiliar areas.
- Depending on others for mobility.

Lucas was selected as the primary persona because he represents the application's main target audience: visually impaired users who are autonomous, comfortable with technology, and regularly travel independently. His needs directly influenced key features such as real-time obstacle detection, voice guidance, and navigation assistance.

Secondary Persona – Sarah

Sarah

age: 35
residence: Paris
education: Bachelor
occupation: Administrative Assistant
marital status: Single

My white cane helps me move forward, but I need information about what it cannot detect.

Sarah is completely blind and travels independently every day using her white cane and screen reader. She has extensive experience with assistive technologies and values accessible digital products. She is particularly concerned about obstacles located above ground level that are difficult to detect.

Comfort With Technology

INTERNET

SOFTWARE

MOBILE APPS

SOCIAL NETWORK

Criteria For Success:

Detect obstacles beyond the reach of her cane.
Access all features using screen readers.

Needs

- Precise audio feedback.
- Reliable obstacle detection.
- Hands-free interaction.

Wants

- Voice-controlled navigation.
- Customizable alert sounds.

Values

- Safety.
- Accessibility.
- Reliability.
- Autonomy.

Fears

- Colliding with undetected obstacles.
- Using inaccessible applications.
- Receiving inaccurate guidance.

Sarah represents blind users who rely heavily on assistive technologies. Her profile highlights the importance of screen reader compatibility, audio feedback, and accessibility standards. Considering her needs helped ensure that the application remains inclusive and accessible to a wider audience.

Tertiary Persona – Marie

Marie represents users with limited digital skills. Her profile influenced decisions related to interface simplicity, readability, and ease of use. Designing for Marie helped ensure that the application remains accessible to users regardless of their technological experience.

Marie

age: 70
residence: Paris
education: High School Diploma
occupation: Retired
marital status: Married, with children

I want technology to help me, not make things more complicated.

Marie is a retired woman with age-related visual impairment. She mainly uses her smartphone for calls, messages, and simple applications. While she appreciates tools that improve her independence, she often feels overwhelmed by complex interfaces and advanced settings.

Comfort With Technology

INTERNET

SOFTWARE

MOBILE APPS

SOCIAL NETWORK

Criteria For Success:

Use the application without assistance.
Understand alerts easily.
Feel safer while walking alone.

Needs

- Large buttons and text.
- Simple navigation.
- Clear voice instructions.
- Minimal setup process.

Wants

- One-touch activation.
- Emergency contact shortcut.
- Very simple onboarding tutorial.

Values

- Simplicity.
- Safety.
- Trust.
- Independence.

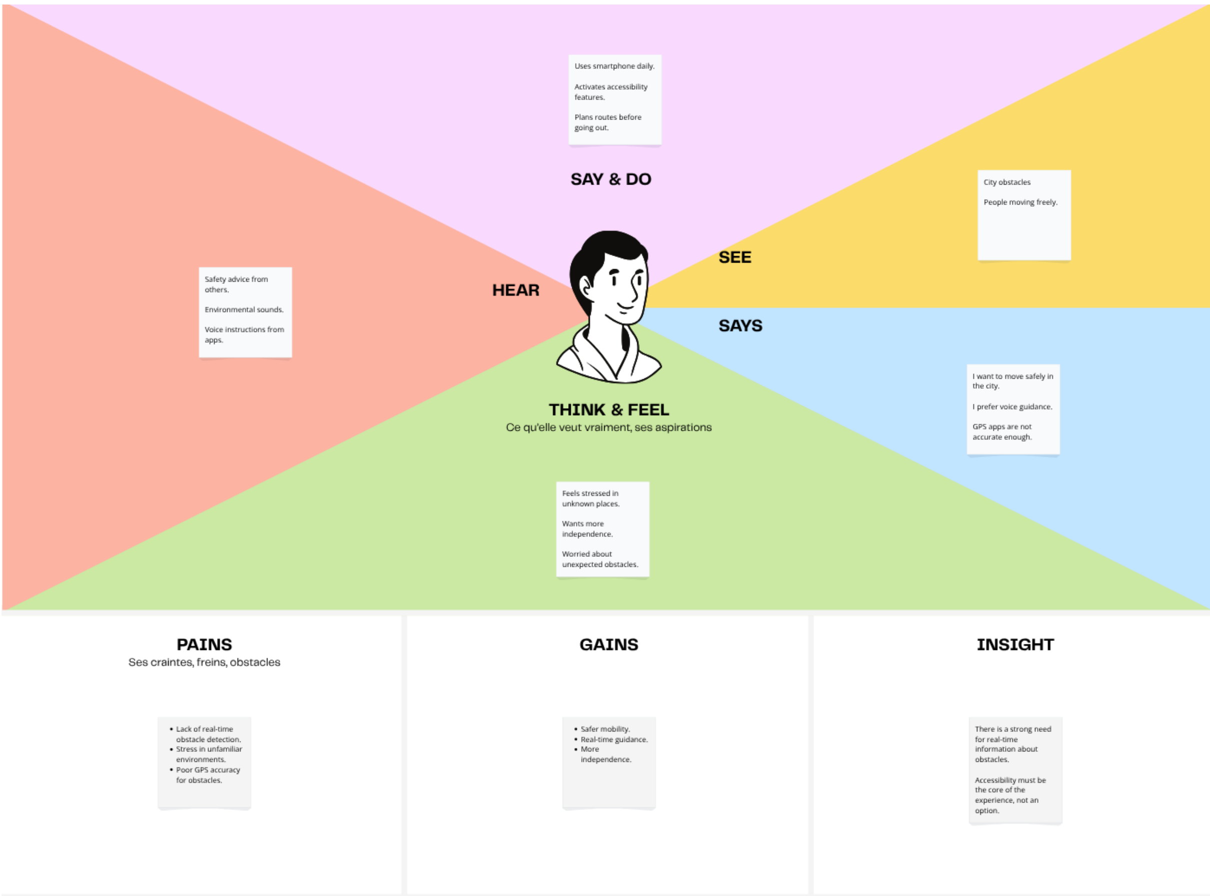
Fears

- Making mistakes while using technology.
- Getting lost.
- Not understanding how the application works.
- Being unable to react to a dangerous situation.

Empathy Maps

Lucas – Primary Persona

This empathy map represents Lucas' thoughts, feelings, behaviors, and needs during his daily mobility. It helps identify key pain points and opportunities to improve accessibility and real-time assistance in the application.



The empathy map highlights that safety, real-time information, and voice guidance are essential for Lucas. It also shows that stress in unfamiliar environments is a major issue. These insights directly influenced the design of the obstacle detection and audio navigation features.

User needs

Safety & Mobility

- Users need to move safely in unfamiliar environments.
- Users need to detect obstacles in real time.
- Users need to be warned about unexpected hazards (scooters, works, signs).

Accessibility

- Users need clear and reliable voice feedback.
- Users need hands-free or low-interaction navigation.

Simplicity

- Users need a simple and intuitive interface.
- Users need minimal steps to access key features.
- Users need easy onboarding without complex setup.

Strategic Analysis

SWOT Analysis

The SWOT analysis was conducted to identify the internal strengths and weaknesses of the application, as well as the external opportunities and threats that may influence its success. This analysis helps define strategic priorities and supports the design decisions made throughout the project.

SWOT map

The SWOT analysis highlights the strong social value of the application, as it addresses a real accessibility need for visually impaired users. The use of voice assistance and AI technologies represents a significant strength, improving both autonomy and safety.

However, the project also faces challenges, particularly regarding technological reliability and ongoing maintenance requirements. Ensuring accurate object recognition and minimizing system errors will be essential for user trust.

The growing focus on digital accessibility and advances in artificial intelligence create valuable opportunities for future development and partnerships. At the same time, competition from existing accessibility solutions and privacy concerns must be carefully considered.

SWOT Conclusion

Based on this analysis, the project should focus on maximizing reliability, accessibility, and ease of use. Future development efforts should prioritize accurate assistance features while exploring partnerships with accessibility organizations to increase adoption and impact.

Study Conclusion

The opportunity study demonstrates a clear need for a mobile application that improves the autonomy and safety of visually impaired users. The personas and empathy maps revealed several daily challenges related to navigation, object recognition, and access to information.

The benchmark analysis showed that existing solutions provide valuable assistance but often focus on specific functionalities rather than offering a comprehensive support system. The SWOT analysis confirmed the potential of the project while highlighting the importance of reliability and accessibility.

Overall, the results of this study validate the relevance of the project and support moving forward to the project planning phase, beginning with the project charter and feasibility study.